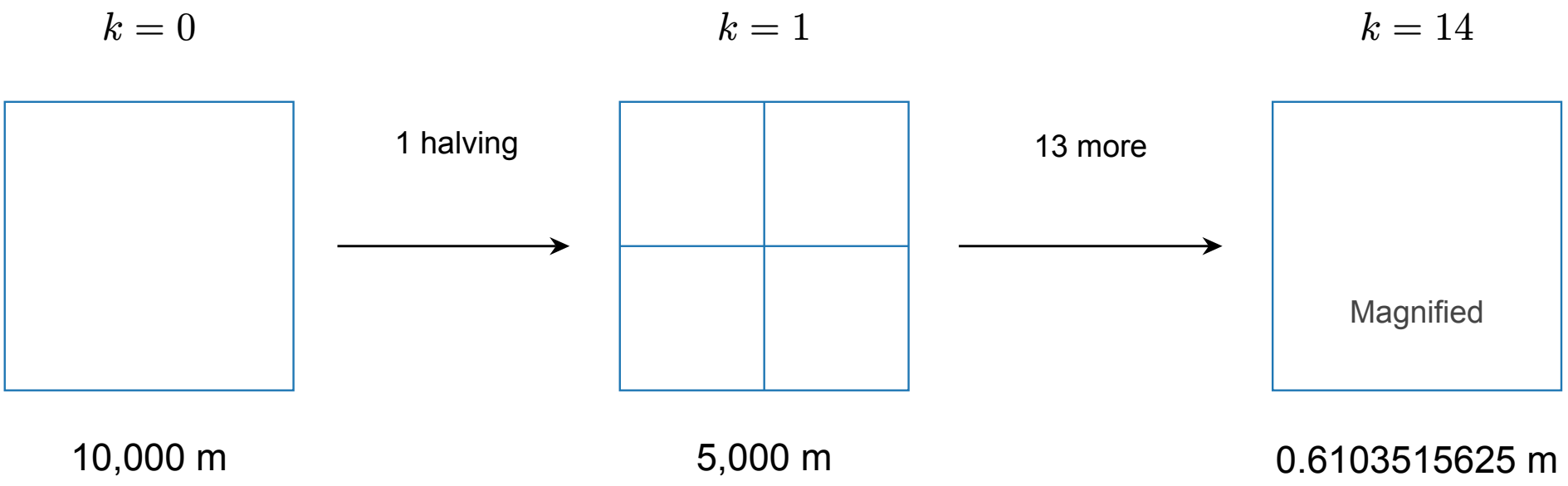

$$h_k = \frac{10000}{2^k}\ \mathrm{m}$$

Mathematical subdivision. Every retained nonterminal cell is split into four.